YOKESH DHANABAL

Master's in AI and Engineering Systems

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Eindhoven, Netherlands | English (IELTS C1) | 10 February 1999 | Orientation Year

SUMMARY

Master's graduate in AI with internship experience at ASML and 2 years of professional experience in a multinational bank. Experienced in building and deploying intelligent systems with a strong focus for production and business impact. I thrive in cross-functional environments, collaborating closely with stakeholders to translate business requirements into scalable, reliable solutions. I love to work as a team and prefer clarity over complexity.

EXPERIENCE

Research Internship (Machine Learning) — ASML

Nov 2024 – Aug 2025 | Veldhoven, Netherlands

- Designed an ML pipeline based on Bayesian Inference for time-series forecasting (reticle deformation) in DUV photo-lithographic scanners.
- Derived and experimentally verified a novel method (called PIPE) to incorporate domain knowledge into the ML model, reducing prediction error (RMSE) by **1.24%** and **5.74%** during normal and state change (clamping) scenarios, respectively.

Team Internship (Deep Learning) — Ambyon

Feb 2024 – Jun 2024 | Eindhoven, Netherlands

- Devised a practical ML solution using LSTM Neural Networks to replace physical sensors on autonomous robots.
- Optimized model architecture for real-time inference, achieving a Mean Absolute Error of **0.29 N (∼1.5% relative error)**, proving the feasibility of deploying the solution in real-time production systems.

Application Developer 01 (Officer C10) — Citibank

Sep 2022 – Jul 2023 | Chennai, India

- **Modernization:** Spearheaded the migration of monolithic views to a responsive Angular Single Page Application (SPA) to improve experience for **10k+** daily users.
- **Automation:** Engineered automated Jira dashboards to streamline project tracking and calculate Defect Injection Rates, saved an estimated **6** hours of manual effort weekly.
- **Reliability:** Worked on end-to-end deployment workflows from local to UAT testing and production labeling and providing critical support to resolve incidents in production.

Application Developer (Officer C09) — Citibank

Jul 2021 – Aug 2022

- **Quality:** Delivered production-ready code for own account transfers module with a defect rate of $<$**0.02** (benchmark 0.15)
- **Compliance:** Implemented accessibility features using semantic HTML and ARIA attributes to meet regulatory requirements (for Hong Kong market), assisting visually challenged users.

EDUCATION

Master of Science in AI and Engr. Systems — TU/e

Sep 2023 – Aug 2025

- Specialization: High-tech Systems and Robotics (Grade: 8/10)
- Relevant Coursework: Deep Learning, Engineering optimization, Computer Vision, Bayesian Machine Learning

Bachelor of Engineering in Electrical and Electronics — PSG Tech

Jun 2016 – Apr 2021

- Completed 1500 hours of industrial training
- Won 11+ academic and extracurricular contests; Served as Secretary and Treasurer of EEE Association, PSGTech.

SKILLS

Python | SQL | C/C++
PyTorch | Scikit-learn | Pandas | NumPy
Azure | Git | CI/CD | Jira
Research work | Report writing | Mentoring
Agile development

AWARDS

- Seethalakshmi Natesan Memorial Award for Best Final Year Project
- Citi Silver Award for securing fourth place in AI OPS Hackathon
- Citi Bronze Award for being the best performing team
- Citi Bravo Award for winning technical Quiz

CONFERENCES

Direct Estimation of Gas-Surface Accommodation Coefficients using supervised ML — *NEFG 2026*

March 2026 | Toulouse, France

HOBBIES

Tennis | Piano